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HOW WE...

How we created virtual patient cases for primary care-based learning

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Introduction

Virtual patient (VP) cases have been defined as 'interactive computer simulations of real-life clinical scenarios for the purpose of medical training, education or assessment' (Ellaway et al. 2006). VP cases are particularly valuable for the construction of clinical reasoning skills, which are notoriously challenging to teach (Balla & Edwards 1986).

We present our preliminary experiences and early application of a selection of video-enriched VP cases designed to meet some of the specific pedagogical challenges of undergraduate education in primary care. The VP cases give students opportunities to practise and refine unique clinical reasoning skills typically deployed by general practitioners (GPs) (Balla & Edwards 1986).

Tailored VP cases have been created for a variety of specialties ranging from geriatrics to paediatrics (Orton & Mulhausen 2008; Gill et al. 2010). Clinical reasoning differs between medical specialities; in particular, GPs draw on unique strategies (Heneghan et al. 2009), which may not be explicitly taught. VP cases can be used to multiply patient encounters providing a bank of verbal and non-verbal experiences (Cook & Triola 2009) which may accelerate acquisition of targeted clinical reasoning skills. VP scenarios are currently used in some areas of primary care education, such as maintenance of certification examinations (American Board of Family Medicine) (Sumner & Hagen 2008) and to quantify variability in primary care physician decision making (e.g. assessing qualified GPs in the treatment of depression in standardised VPs) (Hooper et al. 2008).

Diagnostic strategies employed by GPs have recently been defined (Heneghan et al. 2009), providing some clear learning outcomes to instruct VP case design. Little is reported in the literature about how GPs acquire their clinical reasoning skills, but experience is likely to have an important role (Heneghan et al. 2009). During the consultation, GPs generate prompt diagnostic hypotheses (Elstein et al. 1972). Fast hypothesis generation often depends on prior experience allowing '*spot diagnosis*' or '*pattern recognition*'. We chose to use real patient videos as the basis of our VP cases to provide a virtual 'prior experience' to enable these strategies. Genuine patient videos also offer experience in the interpretation of patient narrative such as a '*classic history*' or patients '*self-reporting*' of a diagnosis which GPs also draw on to make diagnoses (Heneghan et al. 2009).

Practice points

- The unique diagnostic approaches used by GPs tend not to be explicitly taught to medical students.
- VP cases can be tailored using decision-making software to give students experience of primary care specific clinical reasoning skills.
- Collection, over time, of patient videos provides a valuable resource for enrichment of VP cases and should be encouraged.
- Our suggested 'toolkit for success' can be used as a starting point for developing projects which use patient videos.

Video enrichment of VP cases has previously been shown to be beneficial. One study found that users develop enhanced cognitive and meta-cognitive processing when using video-enriched cases compared to text alone (Balslev et al. 2010). The advantages of using video cases to educate non-experts in diagnostic strategies for paediatrics were recently highlighted; patient video cases were found to '*strongly stimulate clinical diagnostic reasoning processes*' (Balslev et al. 2010).

GPs use further strategies such as '*restricted rule-outs*' and '*probabilistic reasoning*' to refine their initial diagnosis (Heneghan et al. 2009). The strategy of '*restricted rule-outs*' involves synthesising a short list of serious diagnoses which must be excluded. Key questions then elicit symptoms suggestive of these diagnoses (*red flags*). For '*probabilistic reasoning*', clinicians estimate a pre-test probability of a patient having a particular pathology and then adjust this probability in the light of test results. Diagnostic decision tree software can be used to create VP cases that give virtual experience of these real-time clinical reasoning techniques.

Follow-up of and reflection on patients is a powerful discipline for the learning and construction of clinical reasoning skills (Kassirer 2010), but this is limited by short

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student rotations. VP cases can be used to compress time, allowing students to observe strategies such as '*test of time*' or '*test of treatment*' which GPs may use to define their diagnosis (Heneghan et al. 2009). This compression of time also allows the evolution of chronic disease, longitudinal management and treatment effects to be illustrated.

On the basis of our preliminary experience and early application of a series of VP cases, we outline our methods for designing and utilising video enriched VP cases and describe how we tailored the cases to promote the acquisition of primary care specific clinical reasoning skills.

How we created the VP cases

VP case selection

We identified key core curriculum topics from the University of Oxford's undergraduate primary care syllabus and selected dyspepsia, low back pain, dysuria and cardiovascular disease for VP case development. In addition to their core nature, these topics were also selected for their amenability for the inclusion of the clinical reasoning strategies outlined above. We do not see this as a comprehensive case list, but rather as a series for pilot study, proof of concept and proof of value.

VP case design

Undergraduate primary care rotations are often short. This means that students are unlikely to observe consultations or meet patients suffering from all of the core curricular presentations to primary care. The aim of our video enriched VP cases is to simulate GP consultations revolving around these core presentations, giving students the opportunity to be active participants in the consultations in a virtual way. Table 1 shows how the various primary care specific diagnostic strategies have guided VP case design.

The VP cases are principally intended for self-directed learning, but could also be used as a teaching tool in a small group or lecture setting. On average, each VP case has 10 questions, comprised of a mixture of true/false questions and multiple best answer questions. The VP cases are linear in design meaning that all students will attempt all questions, but there are optional extra activities e.g. considering a clip of a patient describing their attempts to quit smoking in the cardiovascular disease prevention case. Participants also have the autonomy to repeat video clips or skip them according to their preferences. The average time taken for a student to complete a case is approximately 10 min.

An outline (prototype) for each case was drawn up to include essential content from the syllabus and key elements to a consultation on this topic. Quandary (http://www.halfbakedsoftware.com/) was chosen from among alternative systems for creating and delivering the decision-making scenarios (e.g. Labyrinth – http://labyrinth.mvm.ed.ac.uk/ and SBL Interactive – http://www.sblinteractive.org/) because of its usability and the ease with which it could be customised to create cases. Importantly, it allows a decision tree to be constructed with multiple ways of reaching the same result, with penalties applied for wrong answers.

VP video collection and edition

We utilised recordings from the patient video library of a GP educator (RH) and also made new recordings to fill content gaps, with the advantage that we could pre-plan questions to elicit content-specific responses related to our prototype decision trees. Valid consent was obtained from all patients to use their video for educational purposes within the university virtual learning environment.

Videos were edited (Apple iMovie) and compressed as Flash video, to an appropriate bit rate for web delivery. The appropriate size varies according to the web delivery system used, but most clips were 1–3 MB with a maximum size of 10 MB.

VP case review

Before VP case upload, a review process was undertaken; an independent GP educator reviewed the cases for content accuracy and appropriateness. A member of the university learning technologies team reviewed the technological aspects. Given that the VP cases were intended for self-directed student learning, we asked four students to individually attempt the cases to ensure usability before uploading the cases to the virtual learning environment.

Feedback on the VPs

Feedback was collected on the accessibility, popularity and any perceived disadvantages of each VP case. A free text box (to allow open ended responses) formed the final page of each case. In addition to qualitative feedback, quantitative data were collected on the proportion of students reaching the feedback page (meaning they finished the case) and the number of attempts made at each case per student.

What's next...?

In the light of our experiences, we highlight a number of challenges and future directions for the development of primary care VPs.

Content for VPs

We can speculate intuitively that certain topics, particularly with distinctive visual cues or classic patient narratives, would gain more value from patient video enrichment. Common presenting complaints should be prioritised along with rarer cases which are easily missed without prior experience. There is an unanswered question relating to when video recording of a consultation becomes inappropriate (e.g. recording acutely unwell patients and sensitive topic areas such as sexual and mental health issues), good guidance can be found at http://www.cherri.mvm.ed.ac.uk/cherri.pdf

Quality and availability of patient videos is a limiting factor. We want to encourage GPs to collect patient videos for educational purposes. A bank of patient video clips built up over time, associated with appropriate consent (see below), could be an invaluable resource to enable similar projects.

Table 1. Distribution of diagnostic strategies across the four VP scenarios.

Diagnostic strategy	VP topic			
	Cardiovascular	Dyspepsia	Urinary tract infection	Low back pain
Spot diagnosis	Video clip of patient with hypothyroidism; *overweight, dry skin, wearing many layers of clothing, hoarse voice, slow relaxing reflexes*	Video clip of patient experiencing dyspeptic symptoms and captured belching		Male working in manual labour
Pattern recognition	Video clip of patient with hyperthyroidism following amiodarone treatment; *hot flushes, palpitations, anxiety, weight loss*	Video of patient describing oesophageal pain as opposed to cardiac pain	Patient describing typical UTI symptoms: 'when I go to the toilet it really really hurts, like a sharp pain in my bladder. Its very strong colouring. When I'm working I have to go to the toilet about a million times a day, which is really annoying'	Question relating to pattern of sinister back pains e.g. *cauda equina* syndrome, lateral disc prolapse/root lesion, malignant back pain
Patient self-report	Video clip of patient self-reporting '*I thought I was having a heart attack*'	Patient describing that they were previously diagnosed with a duodenal ulcer	Video clip of patient self-reporting '*I have got one of my urine infections*'	Patient asked 'what do you think might be causing the back pain' Patient responds 'my mum had a slipped disc, so I was worried it might be a hereditary thing'
Classic presenting complaint	Video clip of patient describing myocardial infarction symptoms; *chest pain, crushing, sweating, sense of impending doom. 'I knew something wasn't right'*	Patient describing progressive dysphagia first noticed with liquid and now with solids	Video clips of two different females describing urinary symptoms. Distinguish which patient has the more typical symptoms for a UTI	Patient report 'I was at work. I just couldn't move. It was very tight.' '24/7 I've got an ache' 'its really difficult to sit down for any length of time'
Restricted rule outs	Chest pain; key questions in the history to rule out 'do not miss' differentials. Investigations to rule out MI e.g. *12 h troponin, ECG*	Model questions to rule out sinister causes of dysphagia e.g. *odynophagia, progressive nature, worse with solids than liquids, constant vs. intermittent*	Model sexual history for UTI to rule out sexually transmitted infection	Investigations in back pain e.g. *bone profile, myeloma screen, indications for MRI to exclude disc prolapse*
Red flags	Red flags for cardiac chest pain; *exertional, radiating, associated with nausea, patient reports sense of impending doom*	Dyspepsia as a red flag for malignancy: decisions over who should be referred for endoscopy (NICE guidelines)	Recognition of symptoms are not consistent with a simple UTI e.g. *flank pain, change in vaginal discharge, hallucinations, prolonged symptoms, recurrent UTI*	Red flags for sinister low back pain; e.g. *sphincter disturbance, night pain, leg weakness, previous diagnosis of malignancy.* Video clip model questions to exclude *cauda equina* syndrome
Probabilistic reasoning	ECG showing inferior ST elevation Myocardial infarction to confirm diagnosis	Effect of risk factors on pre-test probability. Identify risk factors e.g. *smoking, caffeine consumption, medications, corticosteroids*	Effect of positive or negative urinary dipstick on probability of UTI	Examination in low back pain and how this effects likelihood of mechanical back pain e.g. *weakness, sensory disturbance, abnormal reflexes*

(*continued*)

Table 1. Continued.

Diagnostic strategy	VP topic			
	Cardiovascular	Dyspepsia	Urinary tract infection	Low back pain
Test of treatment	Patient treated with amiodarone and develops hyperthyroidism; therefore need to change to alternative agent	Knowledge of standard treatment regimens for *Helicobacter pylori*	Video clip of patient 'well I've got a urine infection, but I have had two sets of tablets for two weeks and they haven't worked' Patient then describes symptoms that are not consistent with UTI e.g. bloating, feeling sick, back pain. No urinary symptoms	Video of patient describing his analgesic treatment pathway. 6/52 trial of analgesia
Test of time		NICE guidance – if symptoms persist for more than 6/52 consider referral	Video clip modelling safety netting for symptoms suggesting pyelonephritis	Model explanation of the aetiology of mechanical back pain. Safety netting if symptoms evolve. Follow up if not improved in 6 weeks.
Follow up/referral to secondary care	Patient describes coming into the GP surgery and then being sent straight to A + E after an abnormal ECG in general practice	Test of knowledge of NICE guidelines for referral to secondary care	Recognition of a complicated UTI to be referred e.g. *male, children, recurrent*	Indications for referral e.g. *limb weakness, abnormal neurological examination, bony tenderness*

Content accuracy is another challenge. Concern exists that e-learning resources lack adequate peer-review (Ruiz et al. 2007). We recommend using two separate reviewers to analyse the content and the technological/usability aspects. Previous authors have recommended developing training and guidelines for reviewers (Ruiz et al. 2007), but this is ambitious for a small project. Instead generic guidelines such as those produced by eVIP (http://www.virtualpatients.eu/) can be used in order to standardise quality.

Consent for VPs

A perceived advantage of VPs is their suitability for sharing via the internet (e.g. Jorum/Open Jorum – http://www.jorum.ac.uk/, Merlot-http://www.merlot.org/merlot/index.htm, http://www.virtualpatients.eu/). This pilot project has raised important questions relating to consent for internet access to genuine patient videos.

VP cases are labour intensive to design, and, as such, we envisage the potential for sharing resources across educational institutions, but this is dependent on authors acquiring robust consent. The Common Healthcare Educational Recordings Reusability Infrastructure – Practice, Interoperability and Ethics (CHERRI) project carried out a survey on current practice in sharing clinical recordings and explored the medico-legal context (http://www.cherri.mvm.ed.ac.uk/cherri.pdf). They went on to develop a common consent and licensing model for clinical recordings, which they hope will be widely adopted and is a fail proof approach for future video enriched VP case authors. In addition, the results of the MEDEV open educational resources project will be very helpful in this regard (http://www.medev.ac.uk/oer/).

Evaluation of VPs

During this pilot project, we collected data from three cohorts of students on the 6-week primary care rotation. Approximately one-third of students of all potential students used the cases, giving a total of 82 students who used the cases. We received positive responses with comments, such as '*I liked the real patients!*', '*interactive with patient videos*'. Some commented that they found the scenarios useful for developing consulting skills e.g. '*Good tips e.g. asking if could be pregnant before prescribing*', '*Videos were useful to pick-up how to phrase questions.*'

Patterns of student use give us some insight into how students were interacting with the VPs. Firstly, we observe that students are multiple users; the average number of attempts at each scenario per student is greater than 1, and consecutive attempts occur on different days. This suggests that students may be using the VPs in a formative manner. They can compare their scores on each attempt to map their progress and identify ongoing knowledge/application gaps. Secondly, we observed higher student usage towards the end of their primary care rotation, indicating that they are using the VP cases for revision purposes.

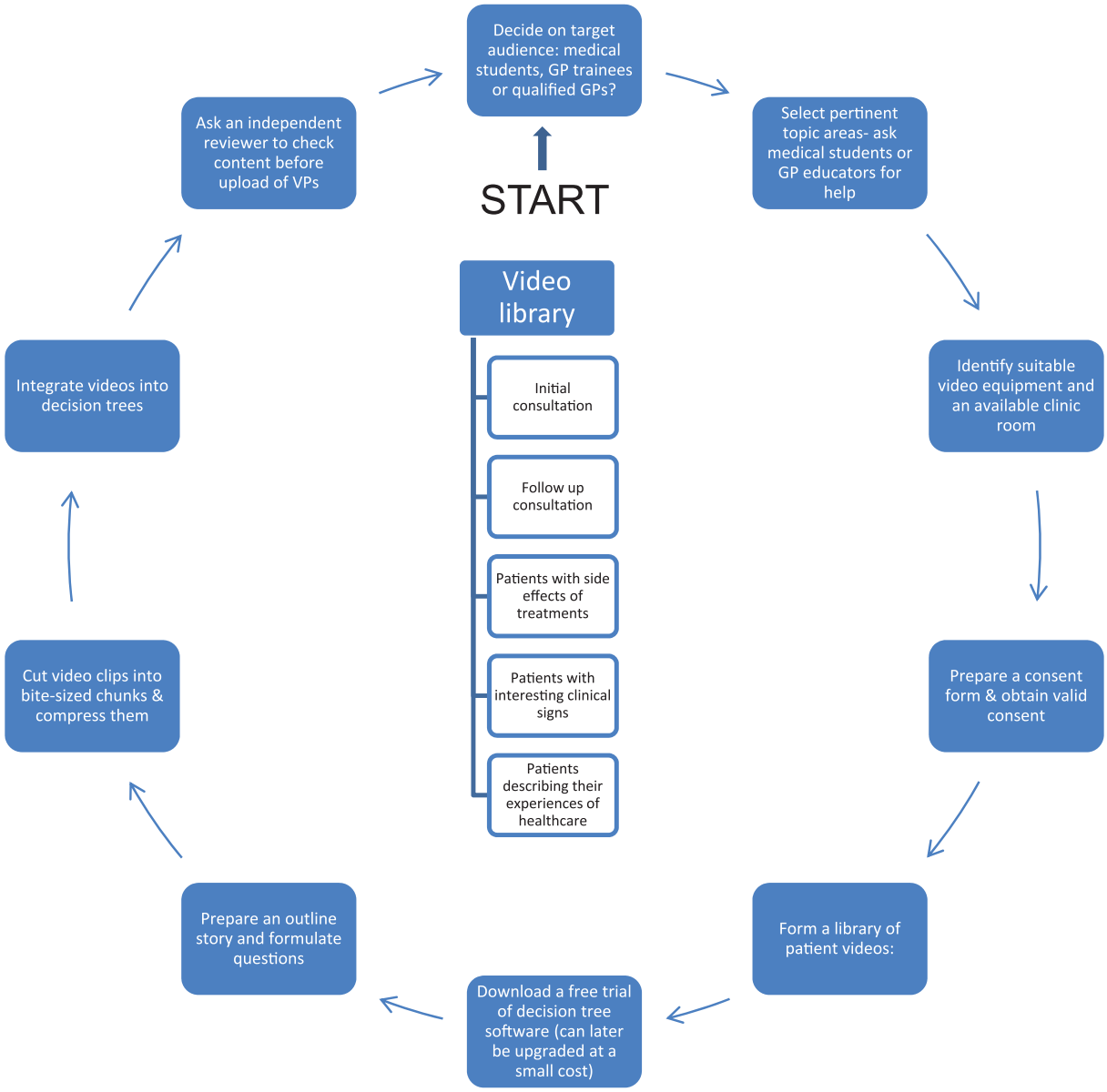

Figure 1. A 'toolkit for success' for those interested in creating similar VPs.

Kirkpatrick defined four levels of evaluation for educational interventions with increasing value (Kirkpatrick & Kirkpatrick, 2006). Our analysis of student attitudes equates to only level 1 feedback. Important further work will involve developing tools for measuring the effect of these cases on student performance (particularly in clinical reasoning skills), student behaviour (particularly relating to diagnostic strategies used) and patient health and satisfaction. It is a priority for us to establish whether the theoretical benefits of VP cases for acquisition of primary care-related clinical reasoning skills can be observed.

Conclusion

We have shown how genuine patient video clips can be collected and integrated into decision-making software. We have used this approach to VP case design to create a learning resource for undergraduate primary care education with the aim of enhancing acquisition of primary care specific clinical reasoning skills. We also believe that this targeted approach to VP case design has wider application. By sharing our experiences, we hope to encourage others to undertake similar projects. By adopting an educated approach to consent and licensing, there is potential for generating a bank of peer-reviewed resources for early development of primary care specific clinical reasoning skills.

A 'toolkit for success'

Drawing on our own experiences in this project, we have produced a 'toolkit' which describes the components of a good video library and the processes involved in turning them from library clip to VP case learning module. We intend this to be used as a brief practical guide for those interested in creating similar VP cases (Figure 1). Centred on a comprehensive video library and using appropriate technological tools, we believe the case can be made for developing VP cases tailored for primary care education.

Declaration of interest: The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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